Lahore University of Management Sciences

AI 500 – Foundations of Artificial Intelligence

Fall 2025

Instructors	Dr. Murtaza Taj
Room No.	Lecture SDSB B2, Office 9-G50A, Gnd Floor, SBASSE Bld.
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Secretary/TA	Check LMS
TA Office Hours	Check LMS
Course URL (if any)	LMS

Course Teaching Methodology (Please mention following details in plain text)
• Teaching Methodology: In-person classes (2.5 - 3 hours meeting once a week) • Attendance is not mandatory as maintaining a good attendance will help students in many ways. Students not frequently attending the lecture will find difficult to cope with the course. We will also have unannounced assessments during the class.

Course Basics				
Credit Hours	3			
Lecture(s)	Nbr of Lec(s) Per Week	1	Duration	2 hours and 30 minutes
Recitation/Lab (per week)	Nbr of Lec(s) Per Week	0	Duration	
Tutorial (per week)	Nbr of Lec(s) Per Week	1	Duration	1 hour (if needed)

Course Distribution	
Core	MS AI
Elective	None
Open for Student Category	MS AI
Close for Student Category	All other students (Open for Auditing)

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COURSE DESCRIPTION
This course offers an in-depth as well as hands-on exploration of the fundamental knowledge needed to excel in the field of data science and artificial intelligence. Aimed at graduate students and industry professionals, this course will server as a foundation for many advance course in the field of machine learning. The course will delve into concepts such as exploratory data analysis, sampling, search, regression, neural network, convolutional neural networks and large language models. This course will offer a blend of mathematical concepts along with hands-on programming examples and assignments. Programming background particularly python skills will be a plus for this course.

COURSE PREREQUISITE(S)	
•	Pre-requisites: None
•	Co-requisites: None

COURSE OBJECTIVES	
	• Foundational Knowledge: Equip students with a robust understanding of calculus behind learning abilities of modern AI algorithms. • Practical Applications: Demonstrate the application of mathematical concepts in real-world ML and AI scenarios, including applications in the domain of remote sensing, medical image, video surveillance etc. • Analytical Thinking: Allow students to design a AI based solution based on the requirements of the applications and its operating conditions.

Grading Distribution: Component Details and weightages
Assignments, 35 % Quizzes, 15 % Mid-Exam, 20 % Final Exam, 30 %

Textbook(s)/Supplementary Readings
Books: • Artificial Intelligence: A Modern Approach by Stuart Russell and Peter Norvig, 4th Edition • Neural Networks and Deep Learning by Michael Nielsen http://neuralnetworksanddeeplearning.com/ Web Resources for Background Material (Python Basics) • Python Tutorial (https://docs.python.org/3.9/tutorial/): Teach yourself python. This is a fairly comprehensive tutorial. • Python + Numpy Tutorial (http://cs231n.github.io/python-numpy-tutorial/) : This tutorial provides an excellent overview of a lot of the functionality we will be using.

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Support Services

LUMS offers a range of academic and other services to support students. These are mentioned below, and you are encouraged to use these in addition to in-class assistance from course staff. For a complete list of campus support services available for you click here (https://advising.lums.edu.pk/#supportservices).

Campus supports & Key university policies

Campus Supports

Students are strongly encouraged to meet course instructors and TA's during office hours for assistance in course-content, understand the course's expectations from enrolled students, etc. Beyond the course, students are also encouraged to use a variety of other resources. (Instructors are also encouraged to refer students to these resources when needed.) These resources include Counseling and Psychological Services/ CAPS (for mental health), LUMS Medical Center/LMC (for physical health), Office of Accessibility & Inclusion/ OAI (for long-term disabilities), advising staff dedicated to supporting and guiding students in each school, online resources (https://advising.lums.edu.pk/advising-resources), etc. To view all support services, their specific role as well as contact information click here (https://advising.lums.edu.pk/#supportservices).

Academic Honesty/Plagiarism

LUMS has zero tolerance for academic dishonesty. Students are responsible for upholding academic integrity. If unsure, refer to the student handbook and consult with instructors/teaching assistants. To check for plagiarism before essay submission, use similarity@lums.edu.pk. Consult the following resources: 1) Academic and Intellectual Integrity (http://surl.li/gpvwb), and 2) Understanding and Avoiding Plagiarism (http://surl.li/gpvwo).

LUMS Academic Accommodations/ Petitions policy

Long-term medical conditions are accommodated through the Office of Accessibility & Inclusion (OAI). Short-term emergencies that impact studies are either handled by the course instructor or Student Support Services (SSS). For more information, please see Missed Instrument or 'Petition' FAQs for students and faculty (https://rb.gy/8sj1h)

LUMS Sexual Harassment Policy

LUMS and this class are a harassment-free zone. No behavior that makes someone uncomfortable or negatively impacts the class or individual's potential will be tolerated. To report sexual harassment experienced or observed in class, please contact me. For further support or to file a complaint, contact OAI at oai@lums.edu.pk or harassment@lums.edu.pk. You may choose to file an informal or formal complaint to put an end to the offending behavior. You can also call their Anti-Harassment helpline at 042-35608877 for advice or concerns. *For more information: Harassment, Bullying & Other Interpersonal Misconduct: Presentation (http://surl.li/gpvwt)*

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AI Policy: Mixed Policy - Selective AI Utilisation

Mixed Policy → Selective AI Utilization

This syllabus statement is useful when you are allowing the use of AI tools for certain purposes, but not for others. Refer to Appendix A for a visual guide to help instructors map AI usage for their assignments.

Template for Course Syllabus

The use of generative AI tools (e.g., ChatGPT, Gemini, Llama, Dall-e, etc.) is permitted in this course for the following activities:

- Brainstorming and refining your ideas,
- Fine-tuning your research questions,
- Finding information on your topic,
- Drafting an outline to organize your thoughts, and
- Checking grammar and style.

However, the use of generative AI tools is not permitted in this course for the following activities:

- Submitting AI-generated content as your own in classroom settings, such as discussion posts, assigned responses, or Zoom chat contributions.
- Completing group work that your group has assigned to you, unless it is mutually agreed upon that you may utilize the tool.
- Writing a draft of a writing assignment.
- Writing entire sentences, paragraphs, or papers to complete class assignments.

These activities are restricted because they undermine individual learning, critical thinking, and the development of essential skills, or misrepresent the student's own contributions to the work. Violations of this policy will be treated as academic misconduct. Please note that AI policies may vary across courses at LUMS, and it is the student's obligation to adhere to the specific requirements of each course.

- Any character in a code/assignment or any other graded instrument not written by you must be properly referenced.
- Use of these tool must not cause obstruction in learning.
- We will conduct vivas and automated checks to ensure you have learned what you have submitted.

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Course Topics and Schedule

Part 1: AI Agents, Search, Classical ML

Week 1-2: Search

- Formalizing actions as a function
- Transition models and their importance
- A* search, Minimax, Alpha/Beta pruning
- **Programming Assignment 1: Search**

Week 3: Unsupervised Learning

- Supervised vs. Unsupervised learning
- K-means Clustering
- Agglomerative Clustering
- **Programming Assignment 2: Unsupervised learning**

Week 4: Supervised Learning and Decision Trees

- Supervised vs. Unsupervised learning
- K-means Clustering, Agglomerative Clustering
- Decision Trees, Bagging
- Random Forest and Boosting

Week 5: Support Vector Machine

- Formulation: Hard SVM and Soft SVM
- Optimization problem formulation (including the concept of duality in optimization)
- Applications and kernel trick
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Part 2: Fundamentals of Data Science with Python

Week 6: Introduction

- Course Overview: Objectives, structure, and expectation
- Introduction to AI and its applications
- What and Why of Data Science
- Data Science Life Cycle
- Intro to Pandas/Python

Week 7: Data Cleaning, Exploratory Data Analysis and Visualization

- Data Sampling (Random, Cluster, Stratified)
- Data Cleaning and Exploratory Data Analysis
- Data Visualization and Transformations
- Case Studies using datasets like crime, election, landslide etc via Pandas/Python
- **Programming Assignment 3**

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Week 8: Mid-term Exam

Week 9-11: Introduction to Causality

- Correlation and causation
- Causal questions: "What if?", "Why?", "How?" Counterfactual thinking
- Real-world problems where causal reasoning matters (e.g., medicine, economics, policy)
- Predictive vs. Causal Questions
- Randomised Control Trials
- Graphical Models & Structural Causal Models (SCMs)
- DAGs as reasoning tools
- Drawing Causal Diagrams (DAGs)
- Confounders, Colliders, Mediators
- **Programming Assignment 4**

Week 12-13: Neural Networks and Back Propagation

- Linear Regression
- Perceptron Model: Binary classifiers, activation functions
- Multilayer Networks: Architecture, forward and backward propagation
- Loss Functions: Cross-entropy, mean squared error
- Optimization and learning using back propagation
- Introduction to tensor flow and/or pytorch
- **Programming Assignment 5: Neural Network**

Week 14: Overview of Advance Topics

- Deep Learning
- Large Language Models
- Agents and Agentic Architectures